# BENEDETTA MARIANI

Computational Neuroscientist · Complex Systems Physicist · Postdoctoral Researcher


| EMAIL | WEBSITE | NATIONALITY |
|---|---|---|
| benedetta.mariani@phd.unipd.it | benedetta-mariani.github.io | Italian |
| **LOCATION** | **DATE OF BIRTH** | **CURRENT AFFILIATION** |
| Padua, Italy | 21 December 1995 | Padova Neuroscience Center |


## PROFESSIONAL PROFILE

**Computational neuroscientist with a PhD in Neuroscience and a background in the physics of complex systems.** Research experience spans dynamical systems, neural data analysis (neural oscillations, neural avalanches, effective connectivity), machine learning and multimodal neural data analysis. Experienced in communicating interdisciplinary research through publications, talks and teaching.

## WORK EXPERIENCE

**Oct 2024 – Oct 2026**

### Postdoctoral Researcher

**Padova Neuroscience Center** · University of Padua, Italy

- Inference of directed brain interactions from fMRI data. Development of predictive models of clinical and behavioral outcomes.
- Supervisors: Prof. Michele Allegra and Prof. Samir Suweis.

**Oct 2020 – Sep 2024**

### PhD Researcher in Neuroscience

**Padova Neuroscience Center / Laboratory of Interdisciplinary Physics** · University of Padua, Italy

- Combined complex-systems physics, computational modelling and neural data analysis to investigate collective behaviour in neural and biological systems.
- Studied how dynamical regimes and oscillatory activity shape computation in neural systems.
- Supervisor: Prof. Samir Suweis.

**Dec 2019 – Sep 2020**

### Research Fellow

**Laboratory of Interdisciplinary Physics** · University of Padua, Italy

- Project: “Criticality in rat somatosensory barrel cortex”.
- Supervisors: Prof. Samir Suweis and Prof. Stefano Vassanelli.

## EDUCATION AND TRAINING

**2020 – 2025**

### PhD in Neuroscience

**University of Padua** · Padua, Italy

- Research conducted at the Padova Neuroscience Center and Laboratory of Interdisciplinary Physics.
- Degree awarded in January 2025.

**Dec 2017 – Oct 2019**

### MSc in Physics of Complex Systems

**University of Turin** · Turin, Italy

- Final grade: 110/110 cum laude.
- Thesis: “Signatures of criticality in rat brain: beyond avalanche statistics”.
- Research performed at the Laboratory of Interdisciplinary Physics and NeuroChip Laboratory, University of Padua.

**Sep 2015 – Dec 2017**

### BSc in Physics

**University of Padua** · Padua, Italy

- Final grade: 100/110.
- Thesis: “Phase separation for intrinsically disordered proteins”.

Oct 2014 – Sep 2015

**Studies in Biomedical Engineering**

**Polytechnic University of Milan** · Milan, Italy

## RESEARCH INTERESTS AND EXPERTISE

**Research areas:** Computational neuroscience; dynamical systems; complex systems; neural oscillations and criticality; neural coding; machine learning for neural data; learning representations in neural data.

**Data and methods:** electrophysiology, neuronal spike trains, functional and effective connectivity, Resting-state fMRI, stochastic and whole-brain modelling, graph and network analysis, statistical inference, machine learning, deep learning.

**Technical tools:** Python (NumPy, SciPy, pandas, scikit-learn, PyTorch, Matplotlib, nilearn/nibabel), MATLAB/SPM, R, C++, Mathematica, Linux/Unix, high-performance computing and Slurm.

## PEER-REVIEWED PUBLICATIONS

2026

**Whisker stimulation reinforces a resting-state network in the barrel cortex: nested oscillations and avalanches**

B. Mariani, R. Guevara, M. Tambaro, M. Maschietto, A. Leparulo, S. Vassanelli, S. Suweis
PLOS Computational Biology.

2023

**Prenatal experience with language shapes the brain**

B. Mariani, G. Nicoletti, G. Barzon, M. C. Ortiz Barajas, M. Shukla, R. Guevara, S. S. Suweis, J. Gervain
Science Advances, 9(47).

2022

**Criticality and network structure drive emergent oscillations in a stochastic whole-brain model**

G. Barzon, G. Nicoletti, B. Mariani, M. Formentin, S. Suweis
Journal of Physics: Complexity, 3(2).

2022

**Disentangling the critical signatures of neural activity**

B. Mariani, G. Nicoletti, M. Bisio, M. Maschietto, S. Vassanelli, S. Suweis
Scientific Reports, 12, 10770.

2021

**Neuronal avalanches across the rat somatosensory barrel cortex and the effect of single-whisker stimulation**

B. Mariani, G. Nicoletti, M. Bisio, M. Maschietto, R. Oboe, A. Leparulo, S. Suweis, S. Vassanelli
Frontiers in Systems Neuroscience, 15.

## PREPRINTS

2026

**Brain-tuning near criticality in newborns by prenatal experience with language**

Jesus Encinas, Benedetta Mariani, Ramon Guevara, Maria Ortiz-Barajas, Judit Gervain, Samir Suweis, Fabrizio Lombardi.

bioRxiv 2026.05.21.726940 · DOI: 10.64898/2026.05.21.726940

2025

**The somatosensory barrel cortex controls the spindle thalamocortical oscillation by frequency locking**

M. Tambaro, R. Guevara, M. Maschietto, A. Leparulo, C. Cecchetto, G. Nicoletti, B. Mariani, S. Suweis, S. Vassanelli. bioRxiv 2025.07.09.662963.

## TEACHING AND SUPERVISION

**Oct 2025**

### Teaching Assistant – Advanced Machine Learning

**MSc in Quantitative and Computational Biosciences** · University of Padua

- Topics: deep feedforward neural networks, convolutional and recurrent neural networks, graph convolutional networks and generative models.
- Module coordinator: Prof. Nicolò Navarin.

**Dec 2022**

### Co-supervisor of BSc thesis in Physics

**University of Padua** · Padua, Italy

- Thesis: "Analysis of neural oscillations and synchrony in the brain".

## SELECTED TALKS AND CONFERENCE CONTRIBUTIONS

**Jun 2026**

### OHBM 2026, Bordeaux, France

Towards consistent and scalable whole-brain effective connectivity estimation.

**Mar 2026**

### COSYNE 2026, Lisbon, Portugal

Self-sustained oscillations link spontaneous and stimulus-driven dynamics in the barrel cortex.

**Oct 2025**

### Bernstein Conference 2025, Frankfurt, Germany

Nested oscillations and critical avalanches prime the rat barrel cortex for whisker stimuli.

**Sep 2025**

### INCTN 2025, Padua, Italy

Poster: Whisker stimulation reinforces a resting-state network in the barrel cortex: nested oscillations and avalanches.

**Jul 2025**

### Annual Computational Neuroscience Meeting (OCNS), Florence, Italy

Spontaneous oscillations and neural avalanches are linked to whisker-stimulation responses in the rat barrel–thalamus circuit.

**Jun 2025**

### StatPhys29 Satellite, Venice, Italy

Contributed talk: Whisker stimulation reinforces a resting-state network in the barrel cortex: nested oscillations and avalanches.

**Feb 2025**

### SIFS Young Seminar Series

Invited talk: Spontaneous oscillations and avalanches are linked to whisker sensory-response processing in the rat thalamocortical circuit.

**May 2024**

### Baby Rhythm ERC Closing Workshop, Padua, Italy

Invited talk: Prenatal experience with language shapes the brain.

**Sep 2022**

### Bernstein Conference 2022, Berlin, Germany

Collective oscillations in the rat barrel–thalamus network; co-author of a second contribution on critical signatures of neural activity.

**Sep 2021**

### Bernstein Conference 2021, online

Disentangling the role of external and intrinsic dynamics in the critical signatures of neural activity.

**Oct 2020**

### Brain Criticality Virtual Meeting

Contributed spotlight talk: Brain criticality beyond resting-state neuronal avalanches.

**Oct 2020**

### Bernstein Conference 2020, online

Brain criticality beyond resting-state neuronal avalanches.

## CONFERENCE ORGANISATION

**Sep 2025** — **Local Organising Committee member**
**Italian Network for Computational and Theoretical Neuroscience (INCTN) 2025** · Padua, Italy

## SELECTED SCHOOLS AND ADDITIONAL TRAINING

**May 2024** — **Stochastic Models and Experiments in Ecology and Biology, Gran Sasso Science Institute, L'Aquila, Italy**

**Mar 2022** — **COSYNE tutorials and main meeting, Lisbon, Portugal**

**Sep 2021** — **EITN Fall School in Computational Neuroscience, Paris, France — project group on mean-field and population models; training in Brian, NEST, NEURON and The Virtual Brain.**

**May–Jun 2021** — **Beg Rohu Summer School, Saint-Pierre-Quiberon, France**

**Feb 2021** — **Computational and Systems Neuroscience (COSYNE), online**

**Dec 2020** — **BrainHack Padova, online — led a project on white-matter microstructure and personality traits using Human Connectome Project data.**

**Sep 2019** — **Summer School in Computational and Theoretical Models in Neuroscience, European Centre for Living Technology, Venice, Italy**

## AWARDS, DISTINCTIONS AND PUBLIC ENGAGEMENT

**2023** — **International media coverage for "Prenatal experience with language shapes the brain"**
The study received coverage from international scientific and general-interest media, including Le Scienze, Nature Italy, El País, The Times, Le Figaro and New Scientist, as well as science podcasts.

**2022** — **Scientific Reports Top 100 Neuroscience papers of 2022**
"Disentangling the critical signatures of neural activity" was included among the journal's Top 100 Neuroscience papers of the year.

**2021** — **SECS Grant**
Scholarship for Events on Complex Systems awarded by the Young Researchers of the Complex Systems Society.

**2014** — **Italian Ministry of Education merit award**
Award for secondary-school graduates achieving the highest final grade with honours.

## LANGUAGE SKILLS

| Language | Level | Certification / notes |
|---|---|---|
| Italian | Native | Mother tongue |
| English | Fluent | Cambridge First Certificate in English |
| German | Basic level | Goethe-Institut, Bremen |